



Low-Cost Remote Water Quality Monitoring (DOpTC)

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Ecology Spring 2027



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Why Monitor Water Quality

- Establishes baseline parameter
- Tracks water system health
- Allows comparison to known standards for drinking water & eco-system
- Alerts to potential pollutants
- Enables faster response time and remedial measures



What is DOpTC

D: Dissolved Oxygen (DO)
Breathable O₂

O: Oxidation Reduction Potential (ORP)
Electron Flow

p: pH
Acidity/Basicity

T: Temperature
Below-Surface (40cm) Temperature

C: Conductivity (Salinity)



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Limitations of Market Options

Historically:

- Cost prohibitive
- Limited customizability or replacements
- Data storage and transmission options are less practical
- Typically limited to in-person single readings



Benefits of DOpTC

- Customizable
 - Add/Subtract Features
 - Replaceable Sensors & Parts
 - Adaptable Housing
- Open source data
- Transmission, storage, and display
- Low Cost
- Long-term data collection
- Solar operation and battery charging



DO Sensor

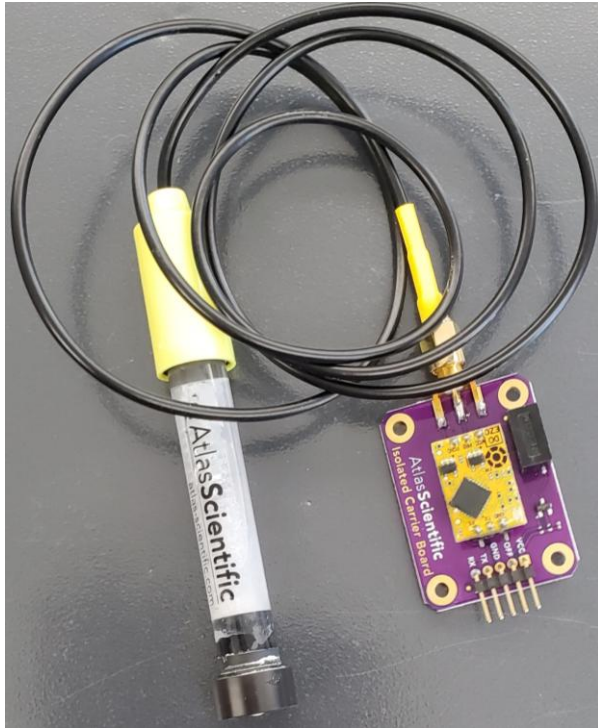
Galvanic Dissolved Oxygen Probe

Range: 0 – 100 mg/L

Accuracy: +/- 0.05 mg/L

Temperature range: 1-60 °C

Annual recalibration



ORP Sensor

Range: -2000mV - 2000mV

Accuracy: +/- 1mV

Temperature range: -5 - 99 °C

Annual recalibration



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pH Sensor

Range: 0-14 pH

Accuracy: +/- 0.002

Temperature range: -5 - 99 °C

Annual recalibration



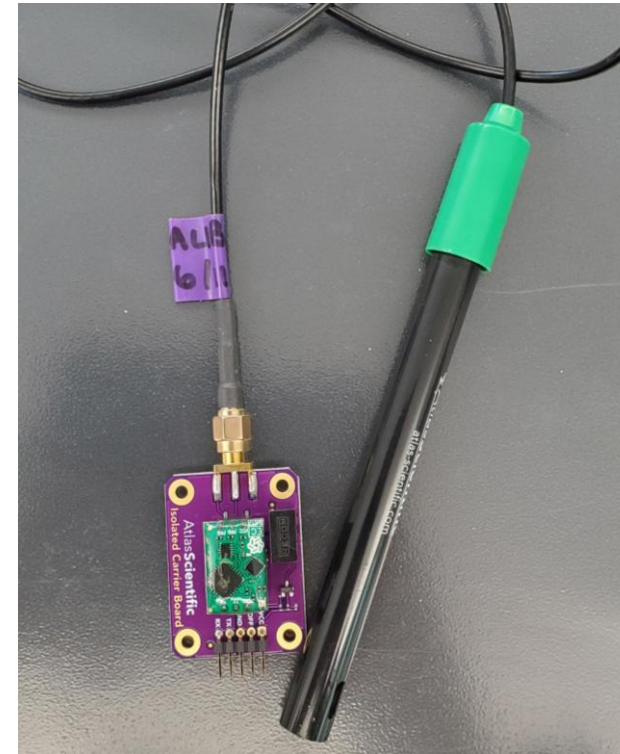
Conductivity Sensor

Range: 0.07 - 50,000 $\mu\text{S}/\text{cm}$

Accuracy: +/- 2%

Temperature range: 1 - 110 °C

Lifetime calibration



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Temperature Sensor

Adafruit DS18B20 Digital
Temperature Sensor

Not for use in salt water or corrosive
environments

Range: -55 – 125 deg C

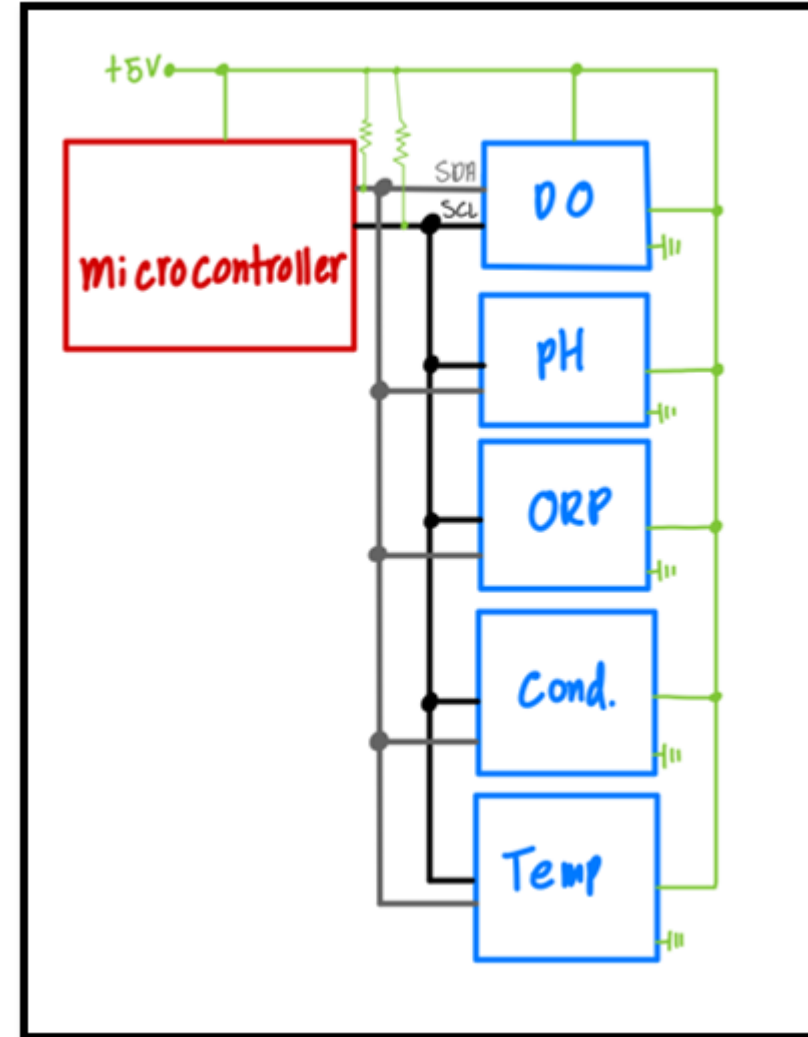
Accuracy: +/- 0.5 from -10 to 85 °C

Dallas 1-Wire

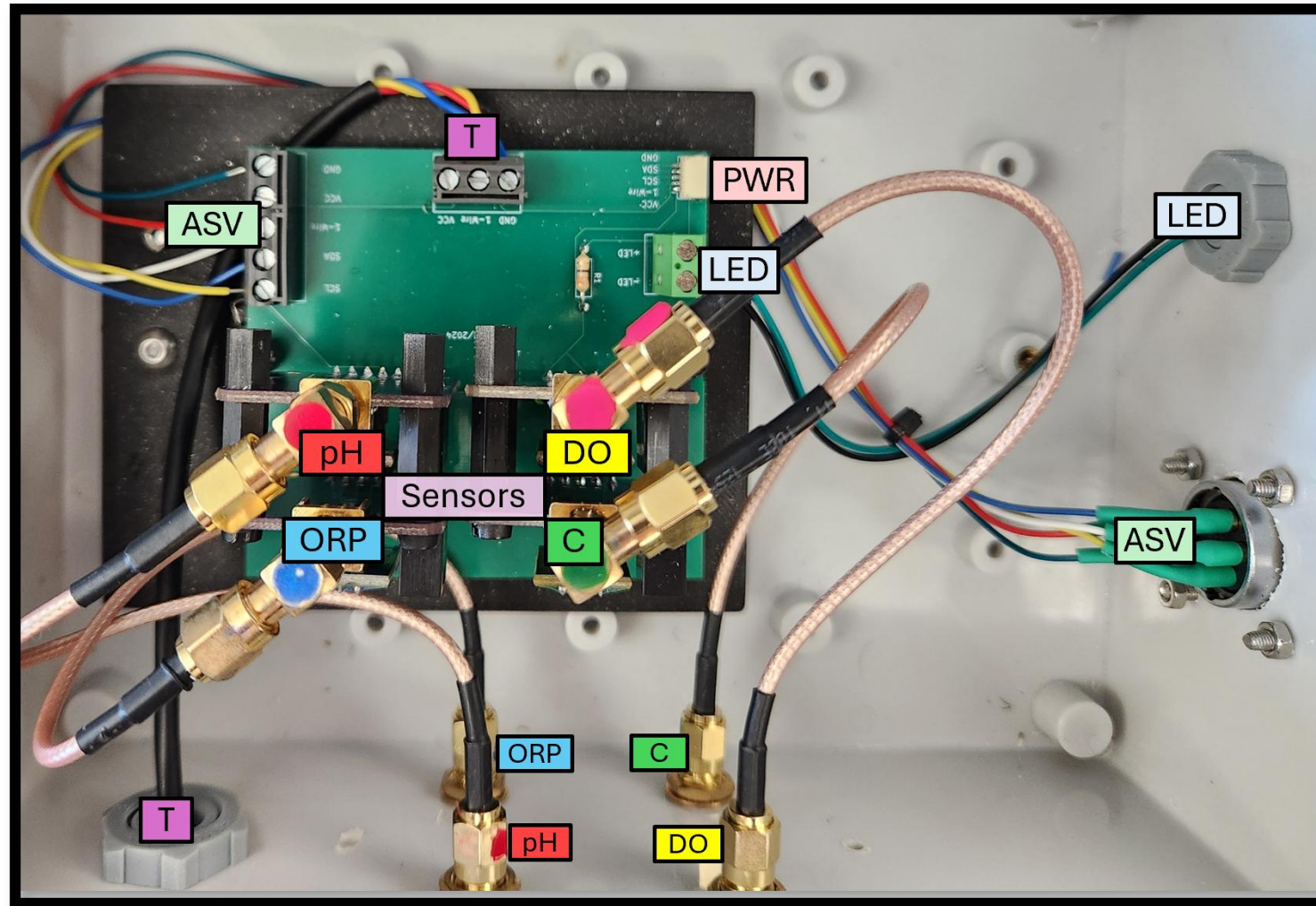


How DOpTC Works

- Sensors Interfaced to a Microcontroller
- I2C protocol
- Temperature is Dallas one-wire
- Data transmission and SD card storage



How DOpTC Works



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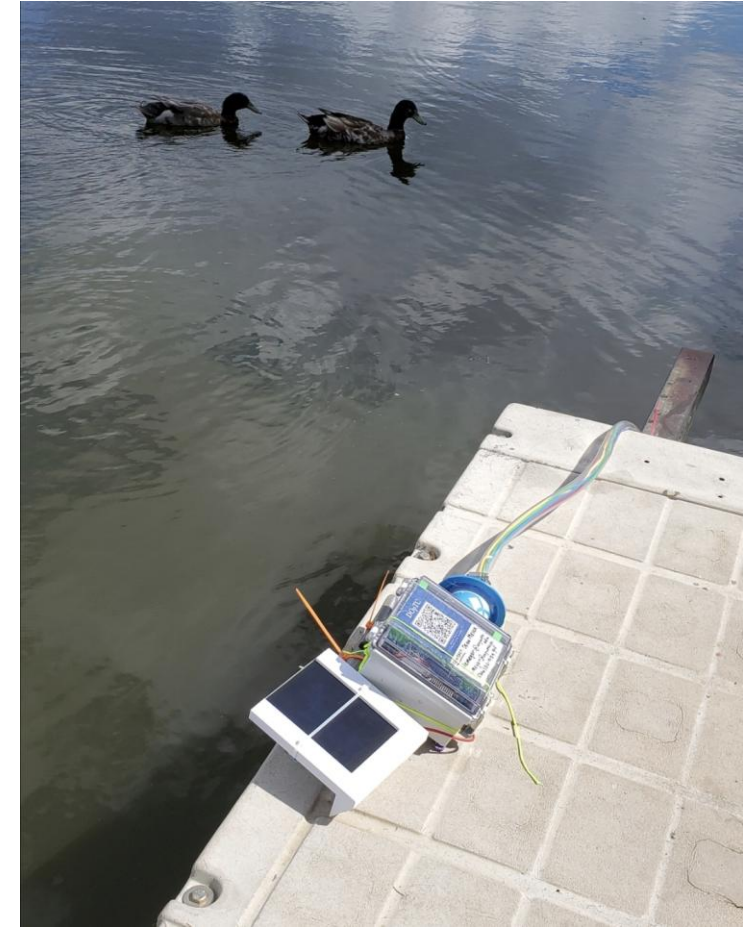
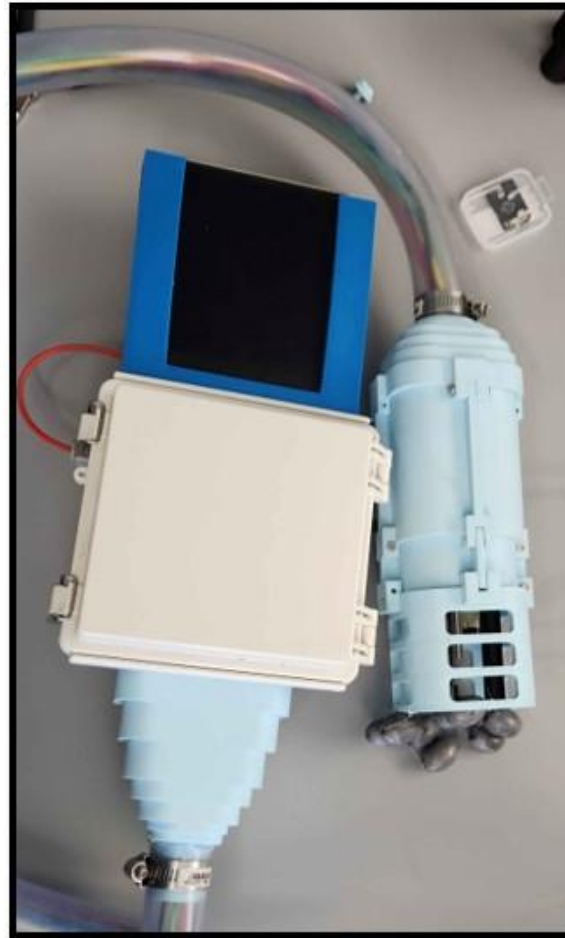
DOP-TC Version 1 – ASV Integration

- Mounts to our autonomous surface vehicle (WIL)
- Data collection along the plotted route
- Writes data to the SD card with GPS tag
- Separate talk on ASV, 10:30 in room G

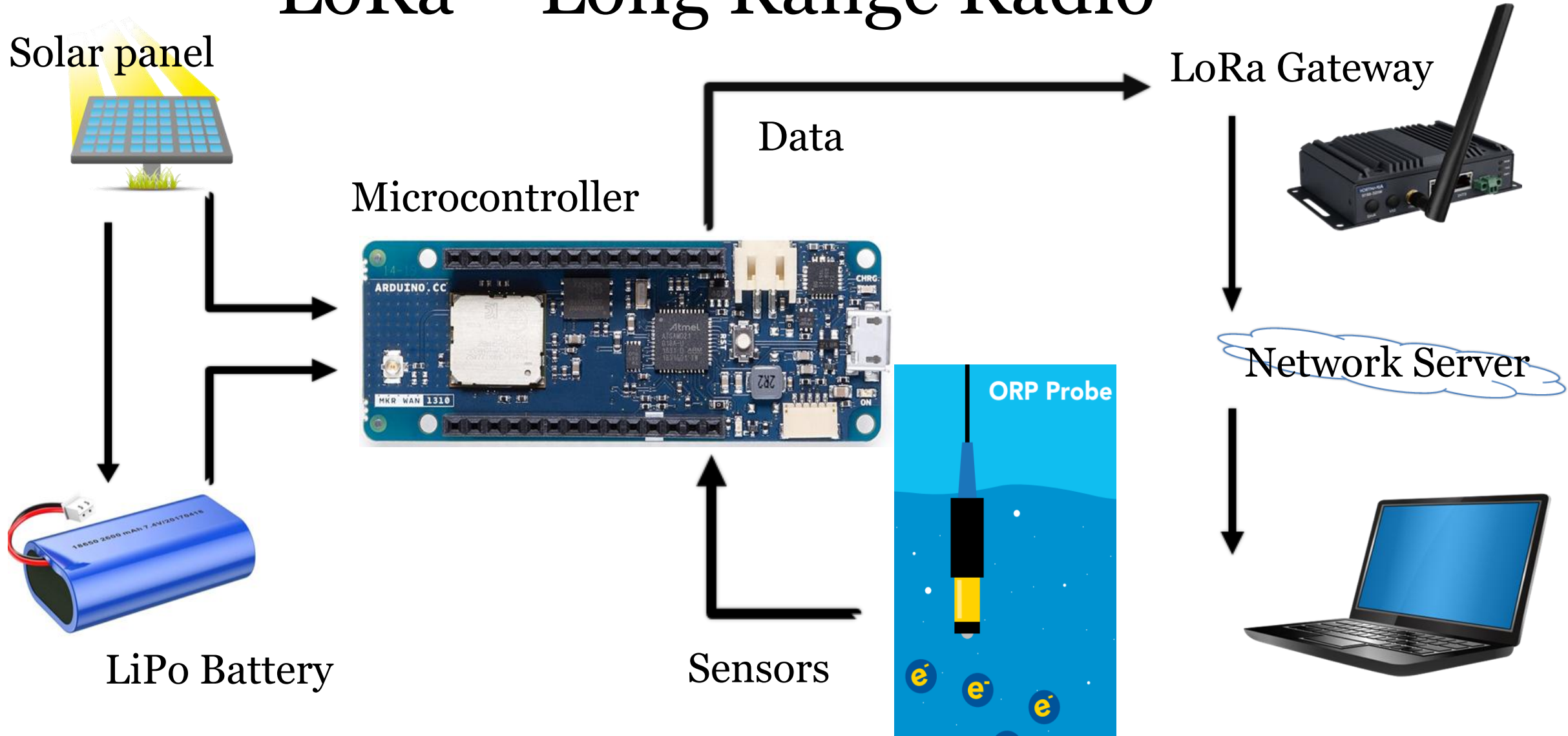


Version 2 – Stand-alone

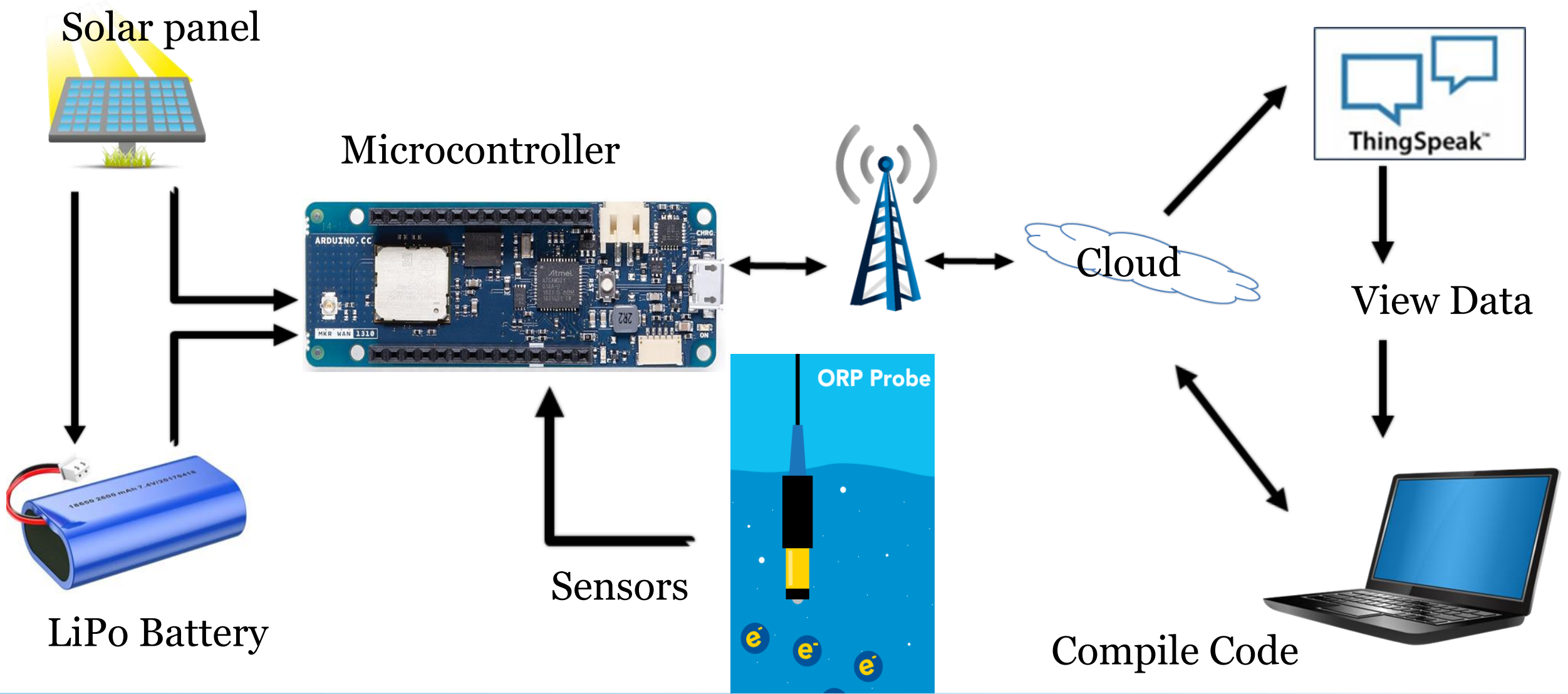
- Cellular or LoRa data transmission
- ThingSpeak data dashboard
 - Timestamped display for each sensor
- Housing redesigned for single-use or long-term deployment



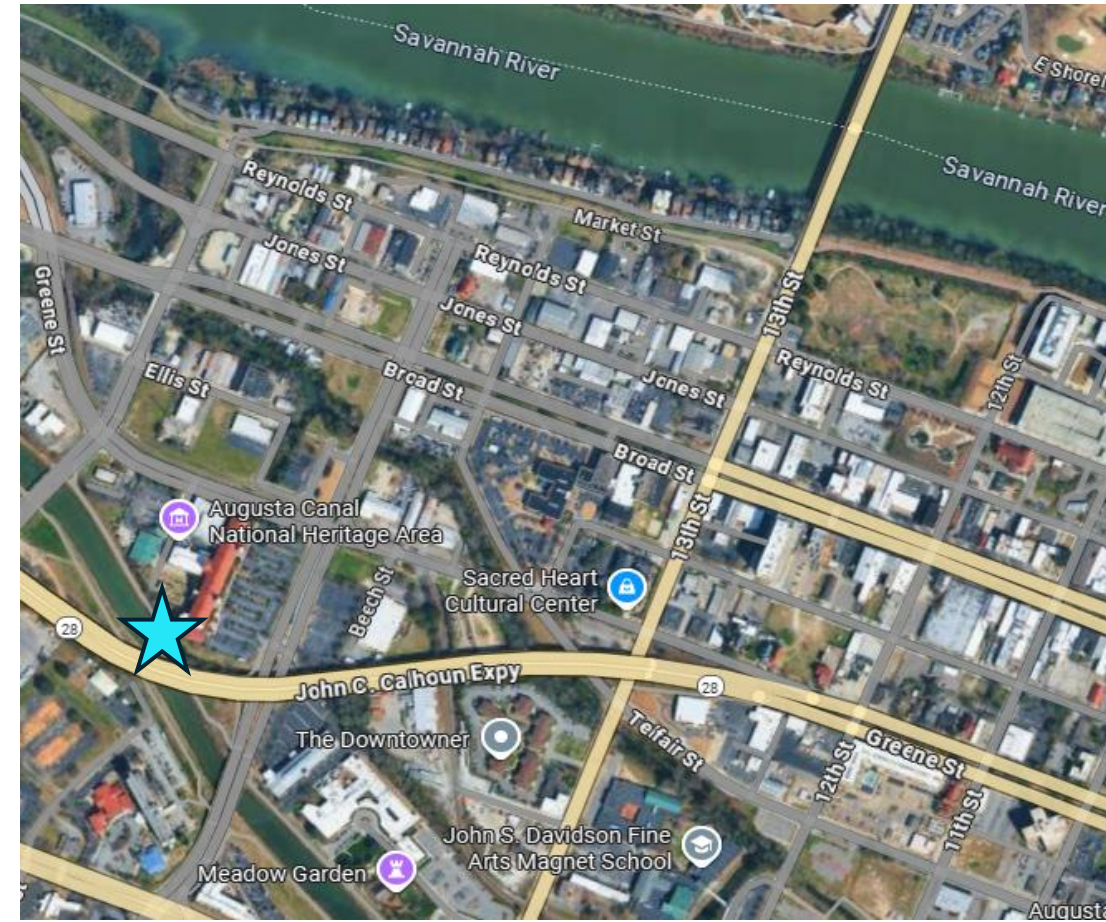
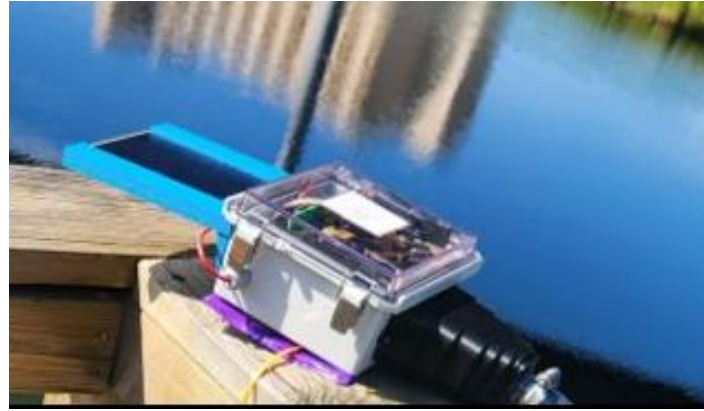
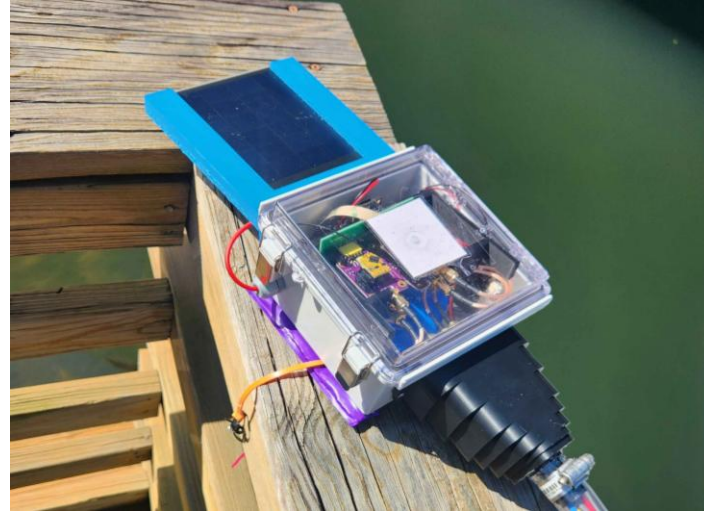
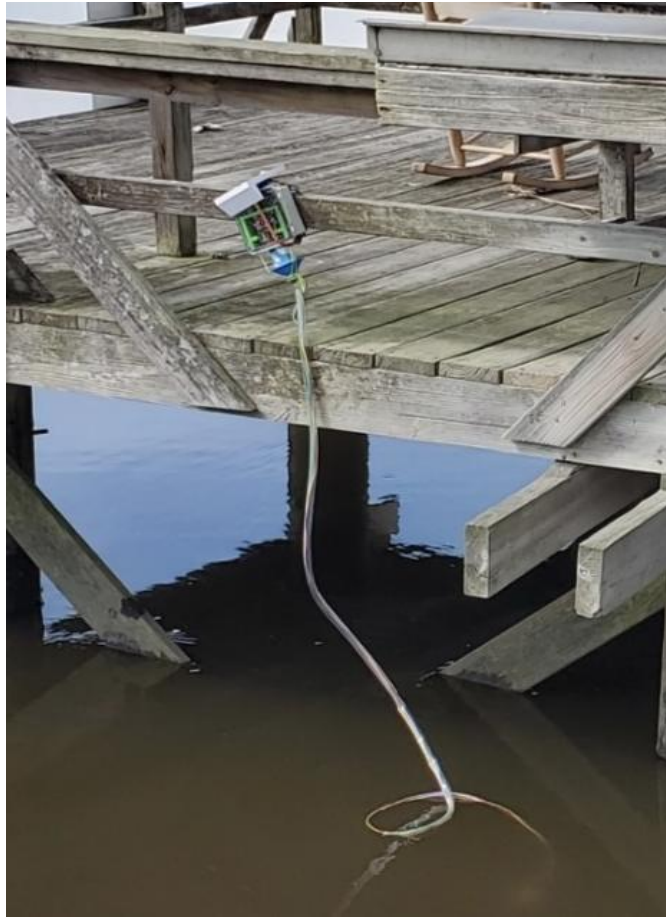
LoRa – Long Range Radio



Particle – Cellular Data Transmission



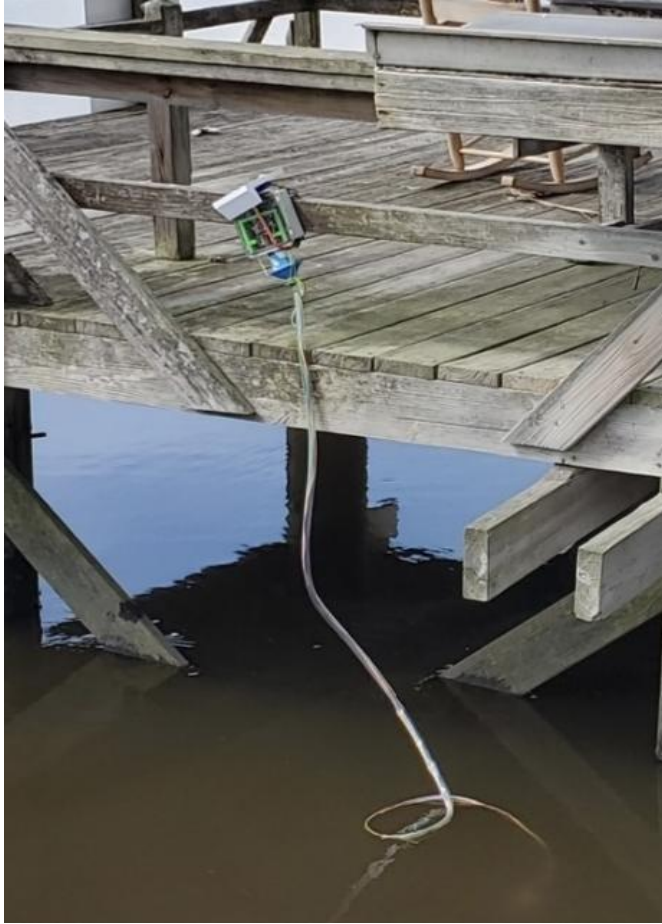
Augusta Canal Deployment



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Satilla River Deployment



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Comparison to Hydrolab HL4

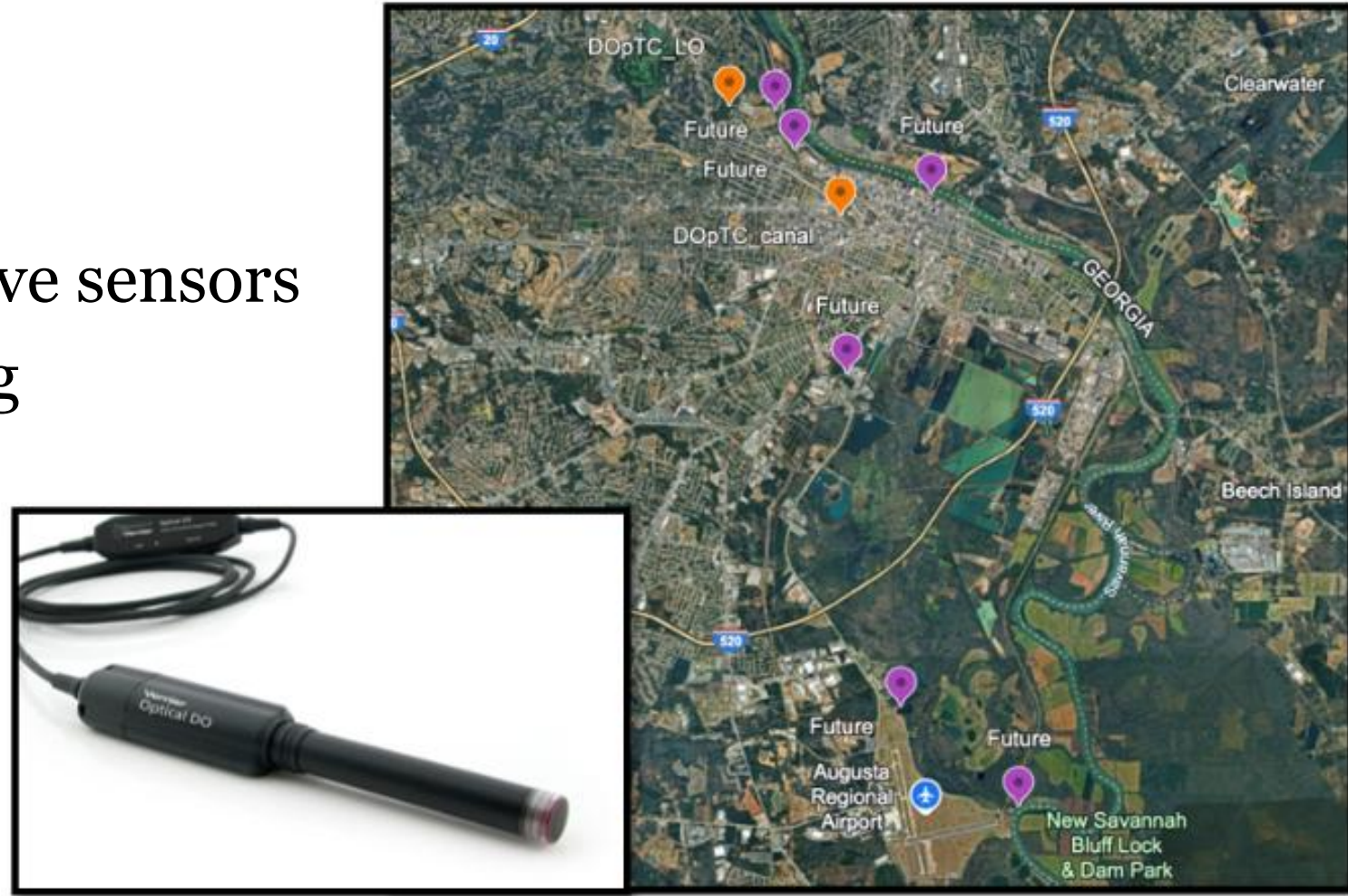
	DO (mg/L)	pH	Cond (µS/cm)	Temp (°C)
DOPC	7.6	7.3	49,188	21.6
Hydrolab HL4	7.6	7.0	48,660	21.5

	DO (mg/L)	pH	Cond (µS/cm)	Temp (°C)
DOPC	7.2	7.2	48,761	22.3
Hydrolab HL4	7.2	6.8	48,721	21.9



Future Plans

- New PCB
- Additional or alternative sensors
- More long-term testing
- Deploy more DOpTCs!



Acknowledgements

- Dr. Joseph Hauger, Augusta University
- Augusta Utilities Department
- Dr. Jessica Reichmuth, Francis Marion University
- Previous collaborators



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Stronger Together

Our research group demonstrates the value of **partnership** between a public utility and a public research university.



Individually, each organization has unique capabilities and knowledge. **Together**, we're able to leverage expertise in chemistry, biology, physics, electronics, and engineering to solve challenging cross-domain problems.

Our **collaboration** so far includes autonomous waterway navigation; monitoring water chemistry, water levels, water velocity, dam seepage, and rainfall levels; and smart city solutions for trash and grass level notifications.